

**A Study of Algorithmic Bias on Inter-sectional Identities: A
Socio-Educational study among Postgraduate Students of
Kanpur city**

SYNOPSIS OF THE DISSERTATION

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1. INTRODUCTION

The world is heavily reliant on machine learning technologies these days. From making court rulings to figuring out little mathematical sums. For the sake of impartiality and excellence. But hold on, are you sure? Nobody's personal prejudice influenced the data you received, and your name was kept private while providing you the solution. In actuality, artificial intelligence (AI) algorithms that claim to improve the precision and efficacy of organizational choices are probably going to reproduce and strengthen societal prejudices ([O'neil, 2016](#)).

Social prejudices in technological terms could be explained by the term, "Algorithmic bias"-it is a socio-technical phenomenon. Its technical facet involves the manifestation of social biases in the results of algorithms, while its social aspect consists of the biases that have been existed in society and affect particular groups, such as marginalized and underprivileged communities for long ([Nima Kordzadeh & Maryam Ghasemaghaei, 2021](#)). It may be possible to mitigate algorithmic bias more successfully if users are aware of how it influences their choices. Additionally, it may alter users' frequency and perception of the algorithms they adopt. As explained by Nima Kordzadeh and Maryam Ghasemaghaei (2021), the technical side of it focuses on how social prejudices present themselves in algorithms' results, while the social aspect includes the long-standing biases in society that harm particular groups, such as marginalized and underprivileged communities. Users' comprehension of algorithmic bias and how it influences their choices may aid in its more efficient mitigation. Additionally, it may alter how consumers view and utilize the algorithms they engage in.

The question now is who is the creator of the algorithms we trust. They might be some of us, from the society we belong. They might have similar views towards a particular minority group in society as other majority groups have. This biased perspective of creators (data analysts and software developers) gives rise to the concept of "intersectionality." Law professor Kimberle Crenshaw is credited with coining the term "intersectionality." A "critical perspective that race, class, gender, sexual orientation, ethnicity, nation, ability, and age function not as unitary, overlapping entities instead as reciprocally building phenomena which in turn shape complex social inequalities" ([Collins, Patricia Hill 2015](#)). In essence, it implies that identity is a complex idea rather than a single entity. Personal identification encompasses a variety of other identities, such as race, gender, socioeconomic status, community, etc. However, intersectionality could be used as a starting point to show how data engineers have approached their work from dominant perspectives, which can be detrimental to social movements that gain traction ([Alicia E. Boyd, 2021](#)).

But, “Identity” is very close definition of self for humans. It is highly individualized and encompasses a person's biological and physical traits as well as social standards. However, identification is crucial for emerging communication technologies in the digital age too. Since you may construct or modify your online persona and alter your actual social identity ([Castells,1997](#)). It may always be changed to reflect one's own self-perception and desired external perception. When using social media and AI, young people are constantly in a state of self-discovery. Students in higher education spend more time on the internet by viewing pictures uploaded by others. As stated by Swapnil Kamble (2023), it creates a fresh perspective on actual people, society, and culture.

The difference between the identity one holds, and the identity he put at social media is in continuous evolution while engaging with a chat-bot or browsing through the social media feed, one keeps constructing a "new algorithmic identity" for themselves. The creation of identities that infers categories of identification from otherwise anonymous entities using mathematical techniques. In the course of defining what gender, class, and race actually mean, it uses statistically significant commonality models to automatically detect a person's gender, class, or race. In the end, it shifts the identifying process to a completely digital and quantifiable level ([John Cheney-Lippold, 2011](#)). An individual's experience in digital space is influenced by their inter sectional identity and this new algorithmic identity.

Apart from all the phenomenons like, algorithmic bias, identity, intersectionality etc. AI has completely transformed human society, altering the paradigms of education, work, and how people engage with technology. Concern and excitement about artificial intelligence's (AI) potential effects on global equities are intricately intertwined ([Crawford 2021](#)). It has the "AI divide," which implies that because there are no differences between researchers, developers, and instructors, it maintains the gaps between those who have the ability to utilize it and those who do not. This aspect of AI is shaped by its developers, whose varied identities and backgrounds play a critical role in determining the course of the technology ([Schiff et al. 2020](#)).

The concern here is the growing integration of technology with conventional educational methods. For a variety of frequently high-stakes applications, such as dropping out predictions ([Christie et al., 2019](#)), digital essays scoring ([Ramineni & Williamson, 2013](#)), obtain degrees applications ([Waters & Miikkulainen, 2014](#)), and skill inference ([Ritter et al., 2016](#)), algorithms have been used extensively in educational practices in recent years. For many years, scholars have been alerting the public to the potential inequalities in educational algorithms' effectiveness and lack of the generalization across communities. Additionally, kids' over-reliance and brain dryness have been raised by this integration.

1.1 NEED AND SIGNIFICANCE OF THE PROBLEM

The rapid advancement of artificial intelligence (AI) applications and our growing reliance on them have altered how we approach problems by applying logic and reality perceptions. Similar to other areas of human development, education has incorporated AI into its theories and tenets. However, as a machine, AI cannot perform tasks as precisely and equitably as people. Numerous studies entirely ignore the qualitative features of AI in favor of concentrating only on its quantitative aspects. The philosophic and inter-subjective facets of artificial intelligence are the main emphasis of this work. how students perceive the impact of algorithmic bias on their academic identities.

Unlike human tutors, AI lacks the empathy and contextual understandings, integrated with algorithmic biases it could affect students' motivation, mental health and the confidence. For eg. There are various schools and ed-tech platforms use AI-driven language learning apps or automated English assessments tools for evaluating students' speaking and writing skills. However, these tools being trained in western English accents and grammar patterns, assume the native accent as mistake. Which ultimately put students in self doubt, loss of confidence and fear of speaking. This might also put inter-sectional bias generating the class divide and group-ism among students.

1.2 STATEMENT OF THE PROBLEM

Numerous studies have been conducted on AI, primarily focusing on its technical aspects. However, little research has been done on its philosophical underpinnings, which include presumptions about intelligence, justice, and ethics. It closes the experiential gap regarding user interaction and perceptions of bias in AI and limits an extensive understanding of its nature and applications.

”A Study of Algorithmic Bias on Inter-sectional Identities: A Socio-Educational Study among Postgraduate Students of Kanpur City”

1.3 OPERATIONAL DEFINITIONS OF KEY TERMS

1.3.1 ALGORITHMIC BIAS:

In this study, Algorithmic Bias means when students interact with artificial intelligence (AI) tools, they may perceive or experience unfair treatment. Such as; a course recommending software might repeatedly recommend “soft skill courses” over technical ones, to a girl from small town, shaping her academic identity.

1.3.2 INTERSECTIONAL IDENTITY:

In this study, Inter-sectional identity means an interconnection of personal and social identity categories that continuously revolve around people's encounters with digital space.

1.4 RESEARCH QUESTIONS :

1. What are the philosophical, social and educational determinants of artificial intelligence?
2. What is the idea of Inter-subjectivity in AI?
3. What are the influence(s) of Algorithmic bias on Inter-sectional Identities of post graduate students in Kanpur city?
4. What are the gender based perceptions on algorithmic bias and its role in shaping their academic identities in Kanpur city?
 - 4.1 What are the perceptions of male postgraduate students on algorithmic bias and its role in shaping their academic identities in Kanpur city?
 - 4.2 What are the perceptions of female postgraduate students on algorithmic bias and its role in shaping their academic identities in Kanpur city?
5. What are the demography based perceptions on algorithmic bias and its role in shaping their academic identities in Kanpur city?
 - 5.1 What are the perceptions of postgraduate students from rural areas on algorithmic bias and its role in shaping their academic identities in Kanpur city?
 - 5.2 What are the perceptions of postgraduate students from rural areas on algorithmic bias and its role in shaping their academic identities in Kanpur city?
6. What are the stream based perceptions on algorithmic bias and its role in shaping their academic identities in Kanpur city?
 - 6.1 What are the perceptions of postgraduate students (M.Ed.) areas on algorithmic bias and its role in shaping their academic identities in Kanpur city?
 - 6.2 What are the perceptions of postgraduate students (M.A.) from rural areas on algorithmic bias and its role in shaping their academic identities in Kanpur city?

1.5 DELIMITATIONS OF THE STUDY

- 1) The sample of this study is confined to postgraduate students of CSJMU university and its affiliated colleges situated in Kanpur City only.
- 2) Only Master of Education (M.Ed.) and Master in Arts, Humanities & Social Sciences (M.A.) will be taken as sample of this present study.

2. REVIEW OF RELATED LITERATURE

2.1 INTERSECTIONALITY

The term “Intersectionality” is a kind of techno-social term by nature, it signifies how the idea of complex identities is, so integrated with algorithms that it act partially with marginalized sections of society in the digital world too.

In the social sciences, intersectionality refers to the notion that an individual's identity is multifaceted and complicated ([Heidi Ledford, 2019](#)). The term "intersectionality" was first used by law intellectual Kimberlé Crenshaw nearly 40 years ago to characterize how Black women were not taken into account by the overlapping social constructs of gender and race in the judiciary ([Kimberle Crenshaw, 1989](#)). According to Patricia Hill Collins (2015), intersectionality offers a "critical perspective that racial or ethnic background, gender, sexual orientation, ethnicity, nation, ability, and age operate not as a single entity but as overlapping entities, that mutually constructing phenomenon that turn to shape multifaceted social inequalities." In the field of computer science, intersectionality is nothing new. It has been utilized in Human-Computer Interaction (HCI) to reconsider the ways in which underprivileged communities have been affected by the design and implementation of technology ([V Yolanda et al., 2020](#)). Furthermore, the concept of intersectionality has been employed as a basis to provide light on how data analysts have been approaching the data science process through dominant perspectives, which can be detrimental to social movements that gain traction ([Alicia E. Boyd, 2021](#)). In many different sectors, intersectionality has been used to emphasize how discrimination and oppression affect individuals differently depending on how their gender, race, and class intersect. It challenges us to examine the phenomenological and material complexity of experience as it is influenced by various overlapping identity traits.

2.2 ALGORITHMIC IDENTITY

The algorithms we interact with continuously while performing some tasks digitally, shape our identity based on the data provided to them and our activities at cyberspace. This continuous interaction often leads to shape the digital behavior of an individual, which is the result of formation of so called; “Algorithmic Identity”

Both the software and hardware components of internet technologies are built with both intended and inadvertent rules that establish the framework that governs our behavior as users in cyberspace. One could refer to these guidelines as "algorithms." Due to the commercial nature of many algorithms, a "new algorithmic identity" is located apart from established liberal politics and civil discourse, but it also enjoys an unparalleled ubiquity in its ability to monitor and store user data. Online, a person's genetics or even how they look do not determine a categorization like gender. It's also not totally self-selected. Instead, individuals are being assigned identity categories depending on how they use the

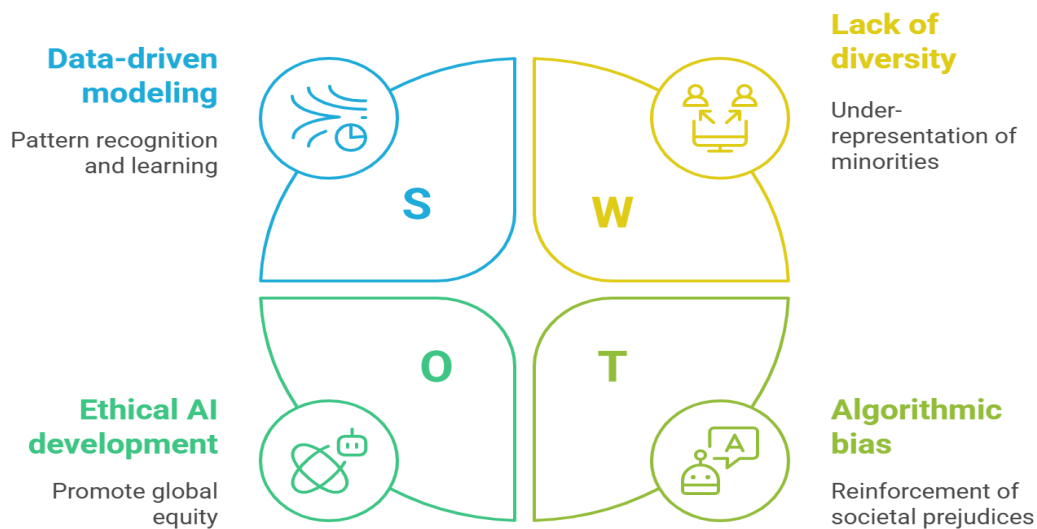
internet to communicate. This is because many algorithms are proprietary, creates a "new algorithmic identity" that is separated from established liberal politics and civil debate ([The theory of Culture & Society,2011](#)). Computer algorithms use mathematical formulas to derive identify categories from otherwise nameless entities. In addition to defining what gender, class, and race actually mean, it uses statistical consistency models to automatically detect a person's gender, class, or race. In the end, it elevates the identifying process to a completely digital and quantifiable level ([2011, Cheney-Lippold,J](#)).

2.3 ALGORITHMIC BIAS

The world is heavily reliant on automated decision making, but as discussed above the concept of Intersectionality and the Algorithmic identity. Both of them play a vital role in assessing the data as well providing the content. However, this data could be biased based on preconceptions involved in the algorithms added by its developers, and behavior of users as well.

By presenting decision makers algorithmically generated information, such as classification and predictive analysis results and the resulting recommendations, algorithmic systems can automate decision-making processes or support human decision-making. According to [O'neil \(2016\)](#), artificial intelligence algorithms that aim to improve the precision and efficacy of organizational decisions are likely to reproduce and strengthen societal prejudices. According to Williams et al. (2018), the concept of bias in algorithms has its origins in social phenomena like social injustice, unfairness, and discrimination. A prominent ethics and a political philosopher John Rawls contends that in a just social structure, people's chances of succeeding in their pursuit of employment and other social roles should be determined by their active skills and willingness to develop them rather than by their background or social class; however, it is morally acceptable to give disadvantaged groups preferential treatment ([Rawls, 2001](#)). It is permissible for an algorithm to disproportionately help poor groups in accordance with Rawls' theory of justice, and thus might not be regarded as an instance of algorithmic prejudice. In conclusion, algorithmic bias can be defined differently depending on the philosophical paradigm used to conceptualize and assess it. This is because different paradigms have different definitions of equality and justice ([Lee et al., 2017](#)). Accordingly, algorithmic bias is a social construct, with definitions that differ among philosophical frameworks and social systems ([Danks & London, 2017](#)).

Algorithmic Bias and Personas in AI



Made with Napkin

2.4 PERSONAS

Personas are the representations of users complex identities. It is more mechanical term as the personas are the result of large data sets and various models involved in interpreting the users identity at digital space. Various studies have explained its concept and its functioning as follows:

Personas are human representations that assist us in addressing design-related queries. They accomplish this by providing us with information about the objectives, passions, and conduct of individuals that are incorporated into their overall persona (J. Grudin, 2006). The majority of methods recommend using user data to build personas. However, data preparation and analysis still need to be done. Usually based on information about the people who are expected to use a technical artifact, challenging personas can occasionally have a theoretical foundation or a special purpose (Jones et al., 2008). The majority of methods advise interacting with actual users, however creating and interacting with personas is subject to prejudices and stereotypes, much like interacting with other people (E. M. Trauth et al., 2016). To ensure the validity of personas, HCI specialists anchor them in objective data (N. Marsden and M. Haag, 2016). In terms of age, color, gender, or sexual orientation, there is frequently an effort to depict user characteristics in a representative manner. However, diversity is lost in this representation (N. Oudshoorn, et al., 2016), which results in a systematic under representation of minority members of the target group: Any attempt to be "representative" results in the elimination of traits which fall in the minority in certain of the aspects deemed crucial (and/or fail to take into

consideration probable impacts), as personas typically arrive in sets of less than 10 personas. By constructing "collective personas" rather than "individual personas" that are meant to represent various aspects of the user group, current approaches to persona construction and use go a little further (A. Giboin, 2011). These "collective personas" are depictions of user collectives, such as communities, groups, etc. Depending on the investigators and how they interpret "collective personas," they are called "group personas," "organizational personas," "persona ecosystems," "communitas," or "collaboration personas." Groups of persons and identifying traits that are shared by these groups are the focus of collective personas. Numerous identities and the complexity that results are frequently overlooked while creating personas, despite the fact that an individual might have numerous group identities (Marsden, N., & Pröbster, M. 2019). Therefore, there has been a growing backlash in recent years against the notion that technology and tech developers are neutral agents devoid of inherent ethical dimensions and values. More individuals are starting to doubt the elevated position that technology holds in society as it gets more pervasive and there are more disturbing tech incidences (such as Google Photo calling Black people apes). (2019, Benjamin Ruha)

2.5 ARTIFICIAL INTELLIGENCE (AI)

Artificial Intelligence is a wide concept to discuss but, this study focuses on its generative functions in the field of education and its role in shaping the diverse identities of students and ultimately shaping their behaviour in real world influencing the confidence as well as motivation level. Below is a brief discussion of studies taken out in similar interest;

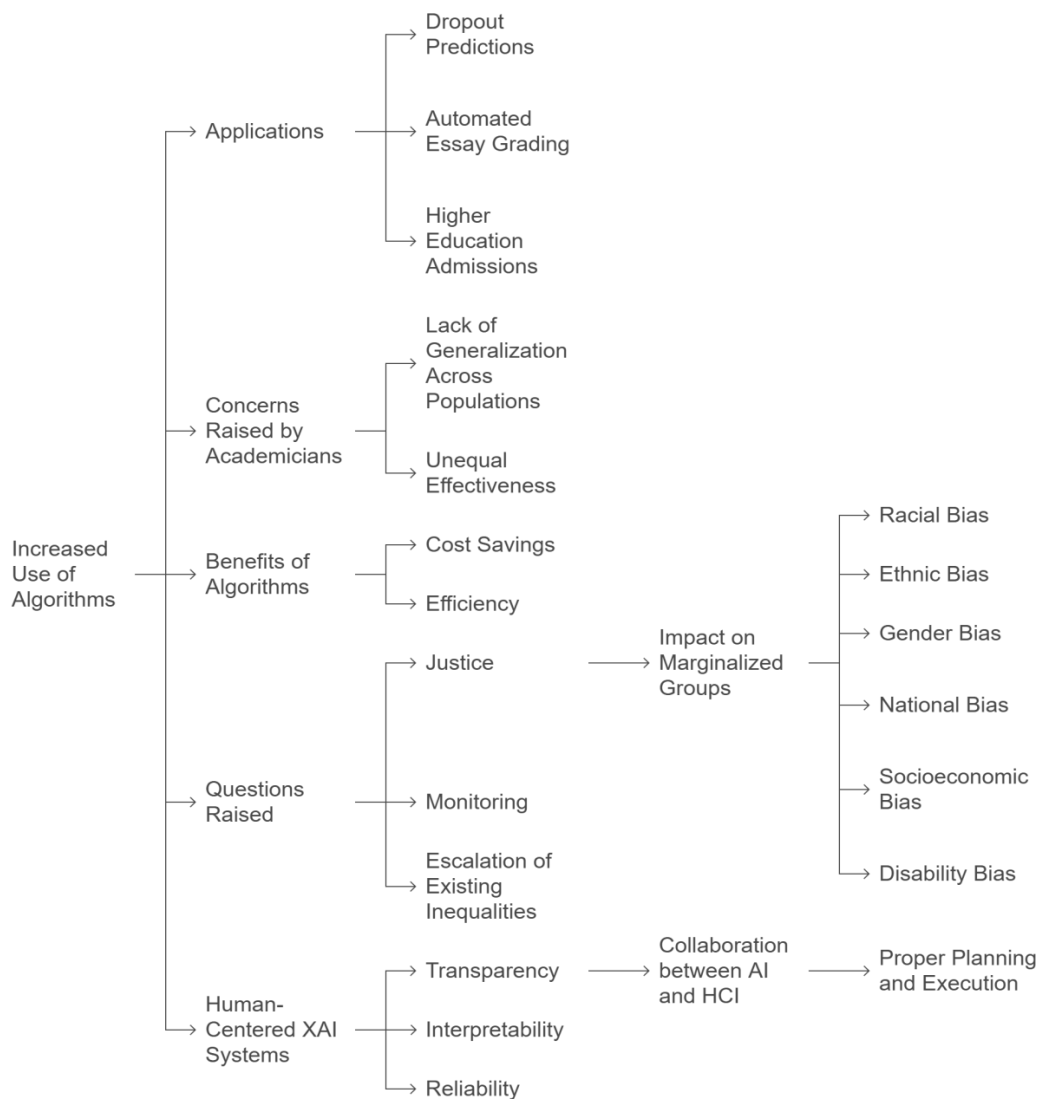
The field of computer science known as machine learning, or AI, is devoted to using data to "teach" computer systems how to model patterns. Regarding AI's potential effects on global equity, there is a complex interplay between optimism and apprehension (Crawford, 2021). The inventors of AI have an impact on its development, and their varied identities and backgrounds play a critical role in determining the course of the technology (Schiff et al. 2020). Human civilization has undergone a revolution, and it has altered how individuals engage with technology, the paradigm of education, and the employment sector. Because of the lack of differences between researchers, developers, and instructors, it has the "AI divide," which implies that it maintains the gaps between those who have the opportunity to utilize it and those who do not. This aspect of AI is shaped by its developers, whose varied identities and backgrounds play a critical role in determining the course of the technology (Schiff et al. 2020).

2.6 ALGORITHMIC BIAS IN EDUCATION

AI has made its space in almost every field , and education is no alien to this trend. Various software s are being used to make education a more interesting and engaging process citing various outcomes, like increased engagement of students and various facilities for prompt feedback and speedy evaluation process. But , citing to the nature of machine languages(algorithms), the academicians have pointed out towards its limitations. Below is the brief discussion relating with the studies taken out in same domain;

Algorithms are playing an increasingly important part in the development of digital learning environments as a result of the growing technological integration with conventional teaching methods. Dropout predictions ([Christie et al., 2019](#)), automate essays grading ([Ramineni & Williamson, 2013](#)), for higher education admissions ([Waters & Miikkulainen, 2014](#)), as well as inference ([Ritter et al., 2016](#)) are just a few of the high-stakes applications for which it is used in educational practices at scale. For a number of years, academicians have been raising warning about educational algorithms' potential lack of generalization across populations and unequal effectiveness. Though algorithms offer cost savings and efficiency, their use raises questions about justice, monitoring, and the escalation of already-existing inequalities ([McConvey & Guha, 2024](#)). Racial, ethnic, gender, national, socioeconomic, and disability-based marginalized groups may be disproportionately impacted by these biases ([Baker & Hawn, 2021](#)).

Algorithmic Bias in Education



We can guarantee the transparency, interpret-ability, and reliability of AI systems by creating human-centered XAI systems. AI and HCI can work with collaboratively to accomplish this. However, for it to work, proper planning and execution are needed ([S. Reddy, 2024](#))

2.7 RESEARCH GAP

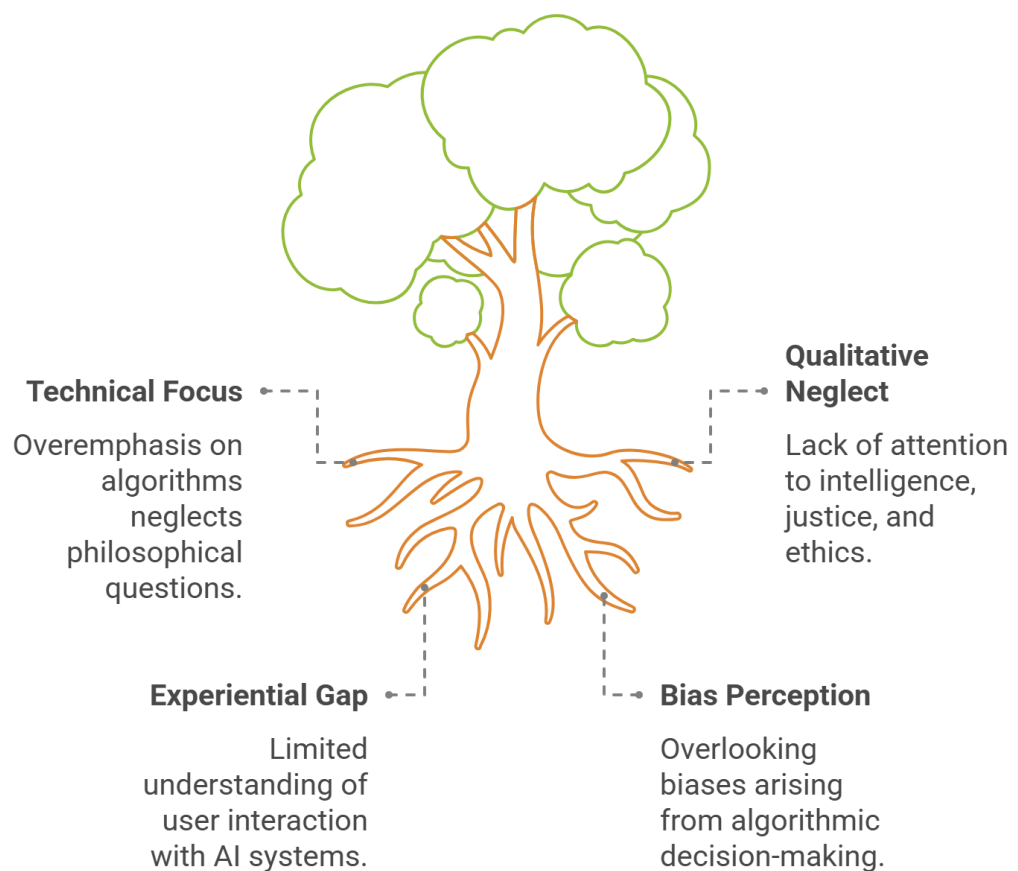
AI has been used as an ice-cake since its invention. It was seen as a paradigm shift in logic and thinking. The developers and academicians were no aliens to it. A range of educational innovations, such as intelligent tutoring systems and assessment and evaluation tools, have been made possible by AI-

powered software's. For lesson plan preparation, curriculum design, content creation, and virtual assistance with administrative tasks and classroom administration.

But there has been a little scope of ethical conceptions. The majority of studies concentrate on its quantitative and mechanical components. There is lack of bias and intellect perception. The major focus has been put on technical aspects of it neglecting the philosophical notions.

Here is a diagram showing the various research gaps observed so far.

Incomplete understanding of AI's role in education due to neglect of qualitative aspects.



3.METHODOLOGY

3.1 APPROACH OF RESEARCH

To investigate the philosophical foundation, inter-subjective and the idea of intersectionality in artificial intelligence (AI) shaped by algorithmic bias, this study will use a qualitative research method. Additionally, the impact of algorithmic bias on Kanpur City postgraduate students' academic identities with survey method.

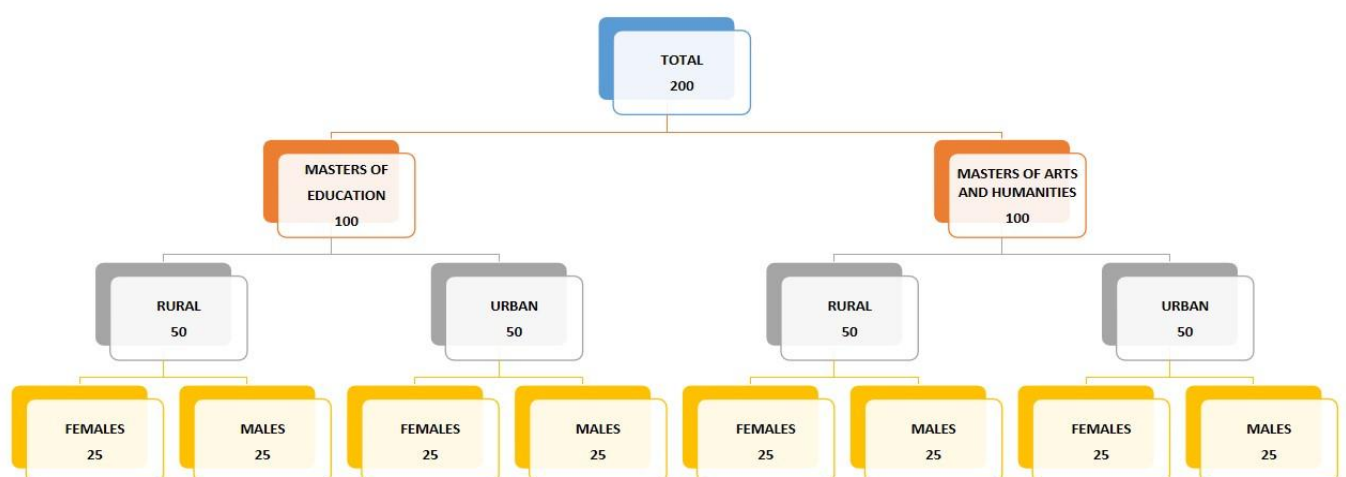
3.2 POPULATION

Population of this study will be

1) Post graduate students and software professionals of Kanpur City, who are engaged with AI-driven academic platforms

3.3 SAMPLE AND TECHNIQUE FOR SAMPLING

1) Up to 200 postgraduate students from various educational institutions of Kanpur City will be chosen using the purposive sampling method. Students who use AI or interact with AI-powered learning tools will be the main focus of the selection process.



3.4 TOOLS FOR COLLECTING DATA

1) Interview : To learn more about how students interpret AI and algorithmic bias, a thorough exploratory interview will be held. Participants will be able to elaborate on their experiences by answering open-ended questions.

2) Self Made Tool: An opinionnaire will be constructed separately for software professionals and post graduate student for collecting their perceptions on Algorithmic Bias.

3.5 DATA TREATMENT PROCEDURE

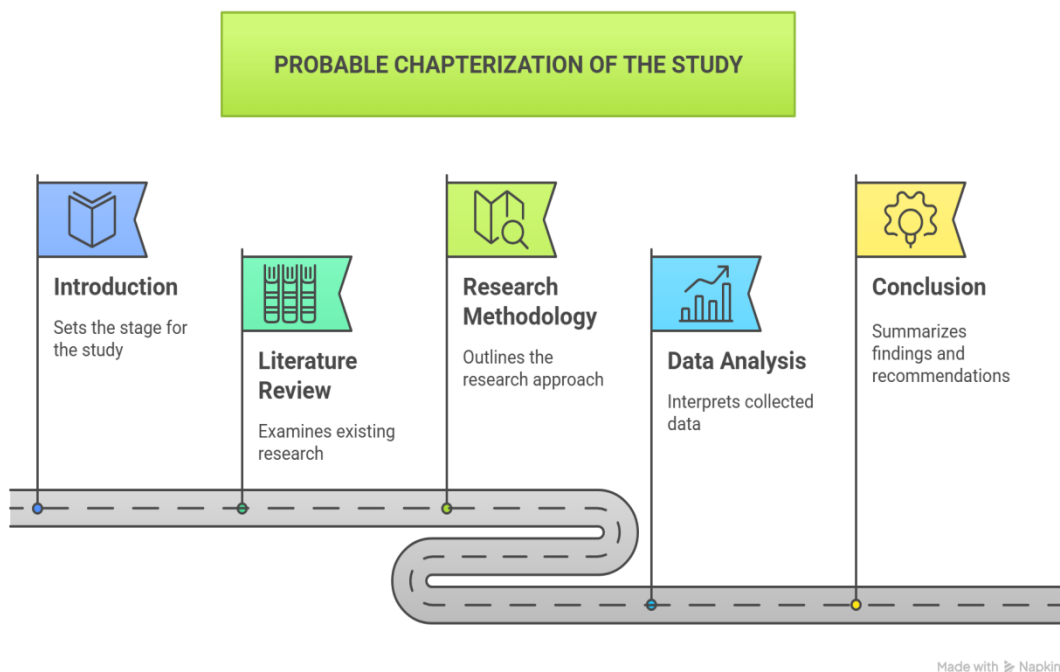
1) Quantitative Analysis: To examine the qualitative data in order to find patterns and topics pertaining to inter-subjectivity and algorithmic bias.

2) Content analysis: To find trends and topics in scholarly journals, research papers, and internet sources.

3) Narrative analysis: To examine the experiences that the participants have in common and find recurring themes and patterns.

4) Transcribe Analysis:

4 . PROBABLE CHAPTERIZATION OF THE STUDY



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